MIL-STD-1312-26
26 JULY 1985

SUPERSEDING
MIL-STD-1312 (In part)
31 MAY 1967

MILITARY STANDARD

FASTENER TEST METHODS

METHOD 26,

STRUCTURAL PANEL FASTENER LAP JOINT SHEAR

FSC 53GP

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DEPARTMENT OF DEFENSE
WASHINGTON, DC 20301

Fastener Test Methods, Method 26, Structural Panel Fastener Lap Joint Shear

MIL-STD-1312-26

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FOREWORD

This standard sets forth a standard test procedure to determine the static strength of sheet metal lap shear joints fastened with quick-release structural panel fasteners.

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1. SCOPE

1.1 Applicability. This method outlines a standard procedure for determining the room temperature static strength of lap shear joints attached with quick-release structural panel fasteners. The significant strength properties include the ultimate strength and the yield strength of the joint. The data is to be used to establish joint allowable loads.

1.1.1 This test method applies to sheet metal lap joints fastened with quick-release structural panel fasteners.

2. REFERENCED DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. Unless otherwise specified, the following specifications, standards and handbooks of the issue listed in the current Department of Defense Index of Specifications and Standards (DoDISS) and the supplement thereto (if applicable), form a part of this standard to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-45662	Calibration System Requirements

HANDBOOKS

MILITARY

MIL-HDBK-5	Metallic Materials and Elements for Aerospace Vehicle Structures

(Copies of specifications, standards, handbooks, drawings and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) forms a part of this specification to the extent specified herein. The issues of the documents which are indicated as DOD adopted shall be the issue in the current DoDISS and the supplement thereto, if applicable.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E 4	Load Verification of Testing Machines
ASTM E 8	Tension Testing of Metallic Materials
ASTM E 83	Verification and Classification of Extensometers

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. DEFINITIONS

3.1 Joint ultimate shear strength. The ultimate strength of a joint is the highest load developed by the joint coincident with or before the failure of the fastener, parent sheet material, or a combination of both.

3.2 Joint yield shear strength. Yield strength of a joint is the load relative to a specified value of permanent set across the joint. The yield shall be determined from the load deflection curve by secondary Modulus Method as illustrated in figure 1 using an offset of 0.04-inch of the nominal stud diameter.

4. GENERAL REQUIREMENTS

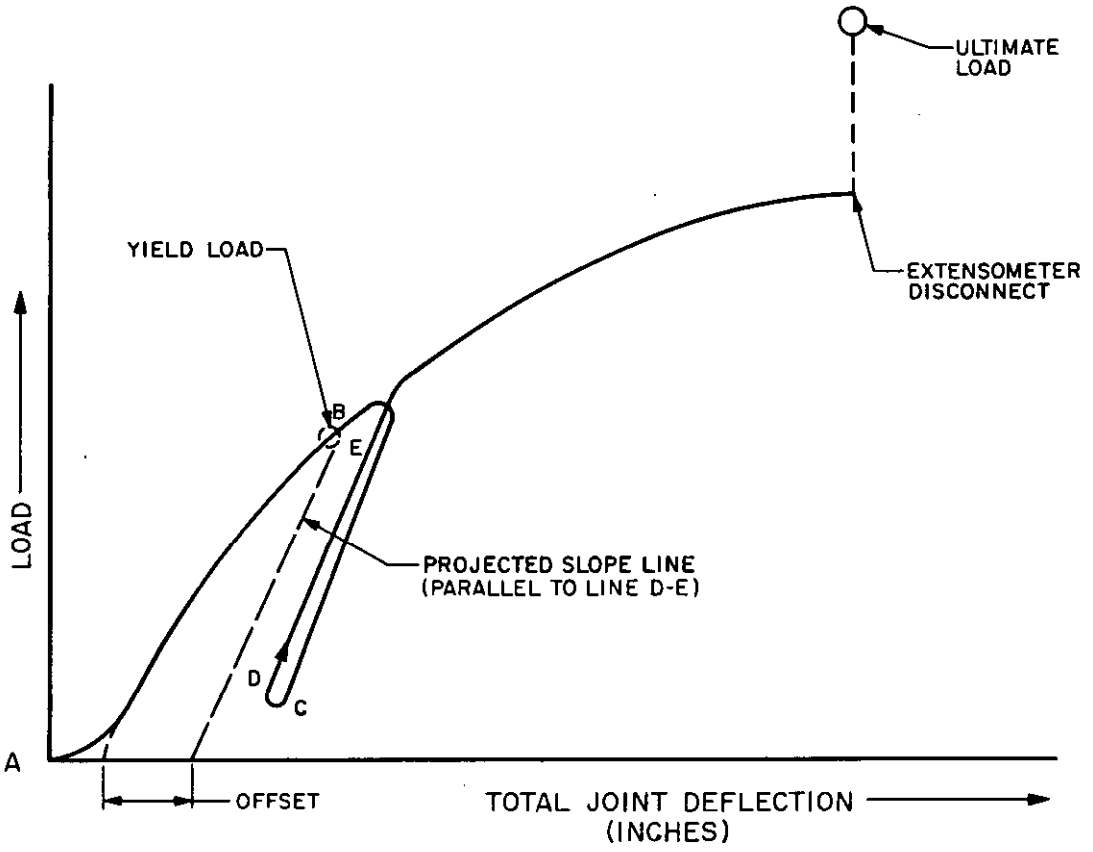
4.1 Test apparatus.

4.1.1 Testing machine. The testing machine shall be capable of applying a tensile load at a controllable rate. The calibration system for the machine shall conform to MIL-STD-45662. Its accuracy shall be verified every 12 months by a method complying with ASTM E 4. The yield loads and the ultimate loads of the fasteners tested shall be within the loading range of the testing machine as defined in ASTM E 4.

4.1.2 Deflection measuring device. The measuring device used shall be an averaging differential transformer extensometer or equivalent, preferably of the separable type. It shall be used in conjunction with an autographic recorder and it shall have an accuracy of 0.5 percent of the indicated joint deflection at yield strength and be calibrated in accordance with ASTM E 83. It shall be capable of direct attachment or mounting to the joint specimen. Load and deflection ranges shall be used which give the initial part of the load extension curve a slope between 45 and 60 degrees.

4.1.3 Test fixtures. The test fixtures holding the joint specimen shall have self-aligning jaws (pin or grip type) capable of transmitting a load along a faying surface of the specimen within 0.005-inch of the centerline of the fasteners.

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Typical load deflection curve for lap joint specimen. Curve shall be developed by loading specimen starting at point "A," to vicinity of approximate yield as noted by point "B." The load shall then be relieved to approximately 20 percent of anticipated load as marked by the path going from "B" to "C." The joint specimen shall be reloaded to define the secondary modulus loop as shown by the curve "C"-"D"-"E." In developing the secondary modulus, selection of load drop-off should be beyond the anticipated joint yield load. After the curve has been sufficiently developed to ensure subsequent determination of the joint yield load, the extensometer may be removed, and the joint loaded to ultimate failure. In establishing the joint yield load from the curve, the slope of the secondary modulus (line connecting points "D" and "E") is projected from the specified offset as established for the specific fastener type and size under evaluation. The point where the projected slope line intersects the load deflection curve is defined as the actual joint yield load.

FIGURE 1. Secondary modulus method for determining yield load deflection curve.

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4.2 Test specimens.

4.2.1 Configuration.

4.2.1.1 Specimen configuration. The test specimen configuration shall be as shown on figure 2.

4.2.1.2 Method for loading. The configuration of the ends of the joint specimen outside the lap area is optional. Friction type holding fixtures, capable of meeting the requirements of 4.1.3 may be used. When the load capacity of the friction grip is limited the use of pin loading holes is recommended. When pin loading holes are used, they shall be located so that the load will pass through the centerline of the fastener holes within 0.005-inch, as indicated in figure 2. When required, doublers as shown on figure 2 are permitted.

4.2.1.3 Edge distance. Standard tests of the lap joint specimens shall be based on specimens maintaining an edge distance of 2D, where "D" = the nominal diameter of the fastener under evaluation. When the effect of edge distance is to be investigated as a variable, it is permissible to change the 2D dimension, but it shall be reported.

4.2.2 Preparation.

4.2.2.1 Joint material. Unless specified otherwise, both elements of the test specimen shall be of the same gage and material. The actual thickness, "T," of each strap shall be measured adjacent to the lap joint area. Dimensions and grain direction of specimen elements shall be in accordance with figure 2. When the effect of grain direction is to be investigated as a variable, it is permissible to use alternate grain directions providing it is so noted in the test report and a drawing accompanies the test report specifically delineating grain direction.

4.2.2.1.1 Joint material mechanical properties. To correlate the data developed under this test procedure, three samples of each parent sheet or plate of material employed in the actual fastener joint strength evaluation shall be tested for tensile strength. Test procedure and method for determination of tensile properties shall be in accordance with ASTM E 8. The grain direction will be the same as required by the material procurement specification.

4.2.2.2 Fastener holes. Fastener holes shall be line drilled perpendicular to the sheet surface with 1°. When countersinking is required, the countersink shall be concentric with the hole within +0.002 FIM. The depth shall be such that the installed flush fastener is flush within +0.002-inch -0.005-inch. Depths for the flush crown head fasteners will be adjusted so that the theoretical sharp shall be flush within +0.002-inch to -0.005-inch. The holes for fasteners shall be relieved on the head side the minimum amount necessary to clear the fastener head-to-shank fillet radius. Recommended fastener hole diameters are listed in

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table I. When functional or fastener requirements dictate, other hole sizes or panel configurations may be specified. The preferred counterbore location for the stud retaining ring is in the lower sheet. Location of the counterbore in the top sheet is optional, but shall be reported. Dimensions of the counterbore shall conform to the applicable standard or manufacturer's requirements. When the test procedure is used for the assessment of joint variables (e.g., gaskets and spaces) specific to a design (not for MIL-HDBK-5 allowables) pertinent details shall be recorded. For example, type of gasket material and geometry (thickness, width and length).

TABLE I. Hole sizes.

NOMINAL HOLE DIAMETER	STUD		SOLID RIVETS 1/	
	MINIMUM	MAXIMUM	MINIMUM	MAXIMUM
0.094			0.097	0.100
0.125			0.129	0.132
0.250	0.25	0.254		
0.313	0.3125	0.3165		
0.375	0.375	0.379		

Note: 1/ Receptacle Attachment Fasteners
2/ Dimensions are in inches

4.2.2.3 Degreasing. Prior to assembly of the test joint, the parent material straps shall be cleaned and degreased.

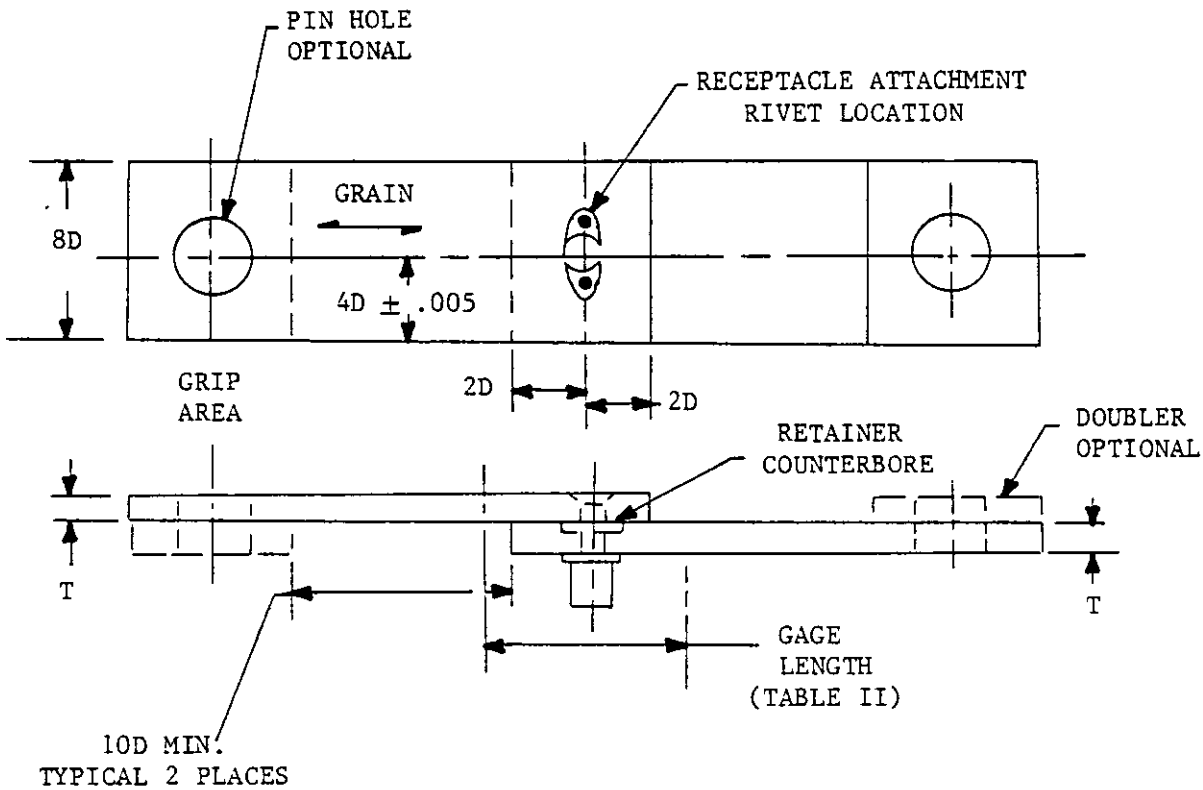
4.2.3 Assembly.

4.2.3.1 Fastener installation. Panel fasteners shall be installed to meet applicable product specification requirements. Receptacle orientation shall be in accordance with figure 2. Studs shall be torqued to the minimum specified value. Test fasteners shall be installed in the "as supplied" condition, i.e., lubrication, except as part of the product specification, shall not be applied to or removed from the manufactured test fastener. Slot-type recessed studs shall be installed with the slot oriented parallel to the transverse axis of the specimen where possible and orientation shall be reported.

4.2.3.2 Grip length. All fasteners tested shall have the correct grip length in accordance with the manufacturer's requirements.

4.2.3.3 Fastener joint preload. Panel fasteners shall be installed with sufficient bearing load on the fastener to ensure that no joint slippage resulting from the fastener clearance will be recorded as joint deflection. The recommended procedure for installation of fasteners is to assemble the straps by tightening the fastener until the faying surfaces are held together loosely and apply the lesser of 100 pounds or 10 percent of the anticipated ultimate joint strength, then tighten to the required torque. Other methods used to provide similar preloads shall be described in the test report.

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NOTES:

1. D = NOMINAL FASTENER DIAMETER
2. T = SHEET OR PLATE THICKNESS
3. FASTENER HOLES (SEE 4.2.2.2)
4. TOLERANCE ON SPACING AND WIDTH = 0.010
5. STANDARD EDGE DISTANCE = 2D
6. COUNTERBORE LOCATION SHOWN IS PREFERRED. LOCATION OF COUNTERBORE IN TOP SHEET IS OPTIONAL.

FIGURE 2. Lap joint test specimen configuration.

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5. DETAIL REQUIREMENTS

5.1 Test procedures.

5.1.1 Installation. The specimen shall be installed in the jaws of the test fixture specified in 4.1.3. The load shall be transmitted along a line passing through the centerline of the faying surface of the specimen within 0.005-inch of the fastener centerline.

5.1.2 Load rate. Unless otherwise specified, the load rate shall be 100,000 ±10,000 pounds per minute per square inch of fastener shear area to the approximate yield load after which it may be increased.

5.1.3 Gage length. For the joint specimen illustrated in figure 2, the specimen gage length across the fastener lap section test area shall be as shown in table II. The deflection measuring device, maintaining the specified gage length, shall be centered across the specimen lap area within 0.063-inch.

TABLE II. Gage length.

NOMINAL FASTENER DIAMETER (INCHES)	GAGE LENGTH (INCHES)
0.250	2
0.313	2
0.375	2

6. NOTES

6.1 Test report. The test report shall include the following data:

a. Sheet material designation and condition. Mechanical properties of sheet tensile specimens.

b. Grain direction.

c. Designation of fasteners used (including complete part number and accessories). Include fastener standards page. Identify lot numbers.

d. Materials and properties of fasteners and accessories when not specified in item c.

e. Measured fastener diameter and hole size. Measured retaining ring counterbore diameter and depth. Counterbore location.

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f. Specimen size including measured thickness, edge distance and nominal width and configuration (figure 2).

g. Model of test machine used and date of calibration.

h. Model of extensometer used and gage length.

i. Loading rate used.

j. Test ultimate load.

k. Recorded load deflection curves. Test yield load, yield offset applied and method of determining.

l. Type of failure in accordance with MIL-HDBK-5, Chapter 9.

m. Report conditions and procedures influencing test data (discrepancies, preload applied, torque value, hole preparation and protective finish used).

6.2 General. The test procedure outlined permits the uniform testing of joints and is designed to eliminate or minimize many of the variables which otherwise might occur as a function of test techniques or method. It is noted that joint strength test data is unique in that it requires special handling and analysis to be properly used and interpreted. The primary purpose of this test standard is to develop and establish load allowable values for design use in accordance with MIL-HDBK-5 or similar other specific application design criteria. Accordingly, where joint strength tests are to be conducted to satisfy the requirements of MIL-HDBK-5, the requirements of 9.4.1 of that document should be observed.

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